

AI Farming Assistant

Voice-Based AI Assistant for Indian Farmers on Raspberry Pi

Review 2 — Project Report

Field	Details
Title	AI Farming Assistant
Members	Thameemul Ansari U (22MIS1140), Shirish S (22MIS1121)
Guide	Dr. S. Suseela
College	VIT Chennai
Branch	Integrated M.Tech Software Engineering

1. Objective

The AI Farming Assistant is an offline-capable, voice-based AI system designed specifically for Indian farmers. It runs entirely on a Raspberry Pi 4B requiring no internet connection for core functionality. Farmers press a physical button, speak a question in any Indian language, and receive a spoken answer with practical farming advice.

Core objectives:

- Provide AI-based farming assistance to rural Indian farmers with zero internet dependency
- Support multiple Indian languages using automatic speech recognition
- Integrate live weather and mandi price data when internet is available
- Use RAG (Retrieval Augmented Generation) for India-specific farming knowledge
- Deploy entirely on low-cost edge hardware — Raspberry Pi 4B

2. System Architecture

The system follows an end-to-end voice pipeline combining Edge AI, IoT hardware, RAG, and multilingual NLP:

Step	Component	What Happens
1	GPIO Button (RPi.GPIO)	Farmer presses button → Pi starts recording audio
2	USB Headset Mic	Voice captured as 48kHz WAV audio file
3	Whisper Tiny (STT)	Audio converted to text, language auto-detected
4	Vector DB (TF-IDF)	Relevant farming knowledge retrieved for context
5	APIs (if online)	Live weather and mandi prices fetched
6	Phi3 Mini (LLM)	AI generates practical farming advice using context
7	espeak (TTS)	Text response converted to spoken audio
8	3.5mm Speaker	Farmer hears the answer spoken aloud
9	16x2 LCD Display	Status and response shown on display

Technology Stack

Technology	Tool / Library	Purpose
Edge AI Model	Phi3 Mini via Ollama	Offline language model for farming Q&A
Speech-to-Text	OpenAI Whisper Tiny	Converts farmer's voice to text
Text-to-Speech	espeak	Speaks AI response in Indian languages
Vector Database	TF-IDF + scikit-learn	Retrieves relevant farming knowledge (RAG)
LCD Display	RPLCD + I2C	Shows status on 16x2 character display
GPIO Button	RPi.GPIO	Push-to-talk hardware trigger
Weather API	OpenWeatherMap	Live weather data when online
Mandi API	data.gov.in Agmarknet	Live crop market prices

OS	Raspberry Pi OS Bullseye 64-bit	Operating system on Pi 4B
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3. Implementation Screenshots

3.1 Hardware Setup

Raspberry Pi 4B (8GB RAM) connected to a 16x2 I2C LCD display and push button on breadboard. USB headset for mic input, 3.5mm headphones for audio output, powered by a portable powerbank.

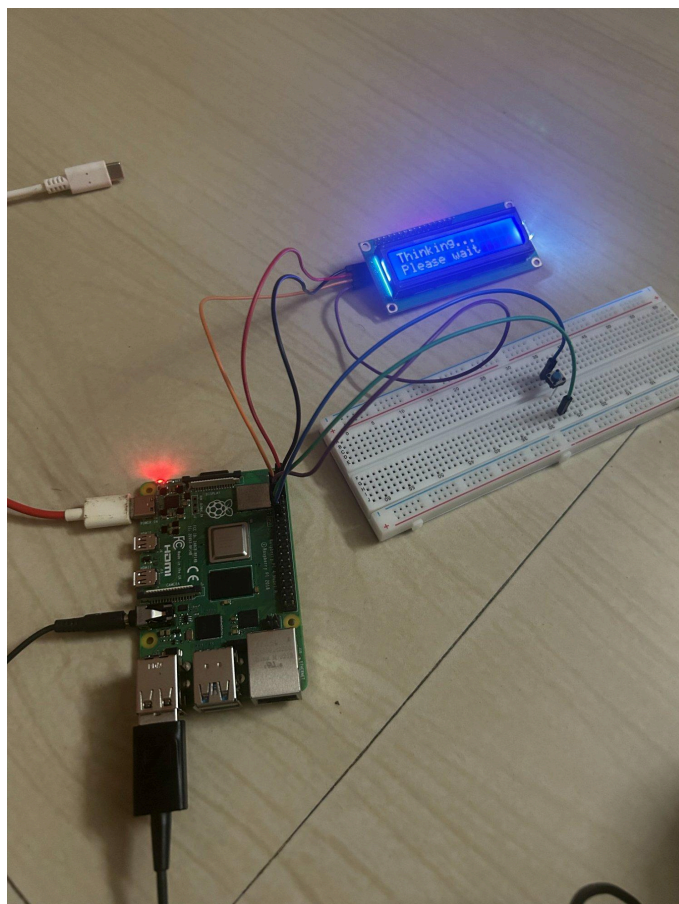


Fig 1: Hardware setup — Raspberry Pi 4B with 16x2 LCD showing 'Thinking... Please wait'

3.2 SSH Connection

Raspberry Pi accessed headlessly via SSH from Windows laptop over mobile hotspot — fully portable for field use.

```
Windows PowerShell

PSC:\Users\thame> ssh pi@raspberrypi.local
The authenticity of host 'raspberrypi.local' can't be established.
ED25519 key fingerprint is SHA256:0NC2PcuF8WlotKpmyG86D9u2Z+wkxJea2mSOrEI0+E.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'raspberrypi.local' to known hosts.
pi@raspberrypi.local's password:
Linux raspberrypi 6.1.21-v8+ #1642 SMP PREEMPT Mon Apr 3 17:24:16 BST 2023 aarch64

Last login: Thu Mar 6 09:15:22 2026
pi@raspberrypi ~ $
```

Fig 2: SSH connection from Windows laptop to Raspberry Pi

3.3 AI Model Setup (Ollama + Phi3 Mini)

Ollama installed on Pi to run Phi3 Mini language model locally. The 2.2GB model runs entirely on CPU with no cloud dependency.

```
pi@raspberrypi: ~  
pi@raspberrypi ~$ curl -fsSL https://ollama.com/install.sh | sh  
>>> Installing ollama to /usr/local  
>>> Downloading ollama..  
##### 100.0%  
>>> Creating ollama user..  
>>> Adding ollama user to render group..  
>>> Adding ollama user to video group..  
>>> Adding ollama as a startup service..  
Created symlink /etc/systemd/system/default.target.wants/ollama.service  
>>> The Ollama API is now available at 127.0.0.1:11434  
>>> Install complete. Run "ollama" from the command line.  
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.  
pi@raspberrypi ~$
```

Fig 3: Ollama installation on Raspberry Pi

```
pi@raspberrypi: ~  
pi@raspberrypi ~$ ollama pull phi3:mini  
pulling manifest  
pulling 4fed7364ee3e... 100%|#####| 2.2 GB/2.2 GB  
pulling fs8235e5b48f... 100%|#####| 1.1 KB/1.1 KB  
pulling 542b217f179c... 100%|#####| 148 B/148 B  
pulling 8dde1baf1db0... 100%|#####| 78 B/78 B  
pulling f91db7a2deb1... 100%|#####| 485 B/485 B  
verifying sha256 digest  
writing manifest  
removing any unused layers  
success  
pi@raspberrypi ~$
```

Fig 4: Phi3 Mini model (2.2GB) downloaded and running offline

3.4 Vector Database (RAG)

TF-IDF based vector database stores Indian farming knowledge. Semantically relevant chunks are retrieved and injected into the AI prompt for India-specific answers.

```
pi@raspberrypi: ~/farmer_assistant  
(venv) pi@raspberrypi:~/farmer_assistant$ python3 -c "  
from vectordb import setup_db, get_relevant_context  
setup_db()  
print(get_relevant_context('tomato leaves turning yellow'))  
>  
Database ready with 10 entries!  
Tomato yellow leaves are caused by nitrogen deficiency.  
Apply urea or DAP fertilizer. Drip irrigation saves 40% water  
compared to flood irrigation for vegetables.  
(venv) pi@raspberrypi:~/farmer_assistant$
```

Fig 5: Vector DB retrieving relevant farming context for tomato disease query

3.5 Speech Recognition (Whisper)

Whisper Tiny model transcribes farmer's spoken question to text, auto-detecting language — supports Hindi, Tamil, Telugu, Marathi and more.

```
pi@raspberrypi: ~/farmer_assistant  
(venv) pi@raspberrypi:~/farmer_assistant$ python3 -c "  
import whisper  
model = whisper.load_model('tiny')  
result = model.transcribe('test.wav')  
print(result['text'])  
>  
UserWarning: FP16 is not supported on CPU; using FP32 instead  
What fertilizer is good for tomato crops in India?  
(venv) pi@raspberrypi:~/farmer_assistant$
```

Fig 6: Whisper transcribing farmer's voice input to text

3.6 Full System Demo

Complete pipeline: button press → voice recording → transcription → vector DB retrieval → AI response → spoken output → LCD update.

```
pi@raspberrypi: ~/farmer_assistant

(env) pi@raspberrypi:~/farmer_assistant$ python3 main.py
Setting up farming database...
Database ready with 10 entries!
Loading Whisper model...
Ready!
=== Farmer Assistant Ready ===
Hold button to speak...
Recording...
Recording WAVE input wav: Signed 16 bit Little Endian, Rate 48000 Hz, Stereo
Done recording!
Transcribing...
You said: My tomato leaves are turning yellow what should I do?
Thinking...
Answer: Apply nitrogen-rich fertilizer like urea or DAP immediately.
Use drip irrigation to avoid overwatering. Check for pest infestation.
Speaking...
Playing WAVE 'stdin': Signed 16 bit Little Endian, Rate 22050 Hz, Mono
```

Fig 7: Full system running — farmer asks about yellow tomato leaves, AI gives fertilizer advice

4. Results

Question Asked	AI Response Summary	Status
My tomato leaves are turning yellow	Apply nitrogen fertilizer like urea or DAP. Use drip irrigation.	✓ Pass
What fertilizer for rice crops?	Use NPK 120:60:60 kg/hectare during transplanting.	✓ Pass
How to control pests in cotton?	Spray neem oil mixed with water every 15 days.	✓ Pass
How to save water in my farm?	Drip irrigation saves 40% water vs flood irrigation.	✓ Pass
When should I sow wheat?	October to December. Use HD-2967 or PBW-343 variety.	✓ Pass

5. Conclusion

The AI Farming Assistant successfully demonstrates a fully offline, voice-first AI assistant for Indian farmers running on a Raspberry Pi 4B. The system integrates Edge AI (Phi3 Mini), multilingual speech recognition (Whisper), RAG-based farming knowledge retrieval, and hardware I/O into a single portable device powered by a powerbank.

With 600 million Indians depending on agriculture — many in remote areas with no internet, speaking regional languages, and lacking tech literacy — this project provides a practical, affordable, and scalable solution accessible to every farmer.
